Day: Date:

LAB - 1

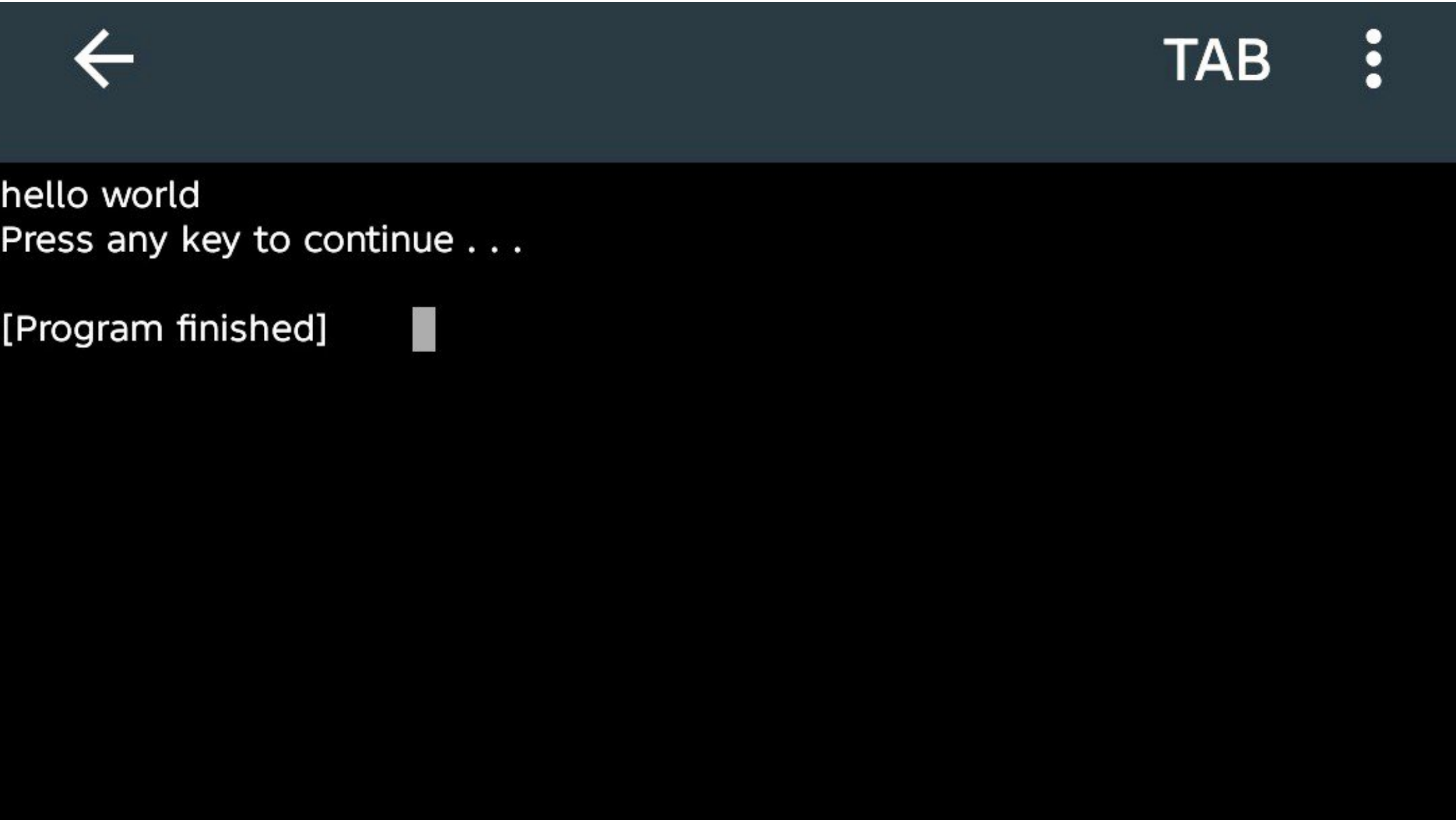
Task 1

Run the following program to get output.

→ input

```
#include <iostream>
using namespace std;
int main()
{
cout << "hello world" << endl;
System ("Pause");
return 0;
}
```

→ · Output hello world


TAB
hello world
Press any key to continue . . .
[Program finished]

Day: _____ Date: _____

Task-2

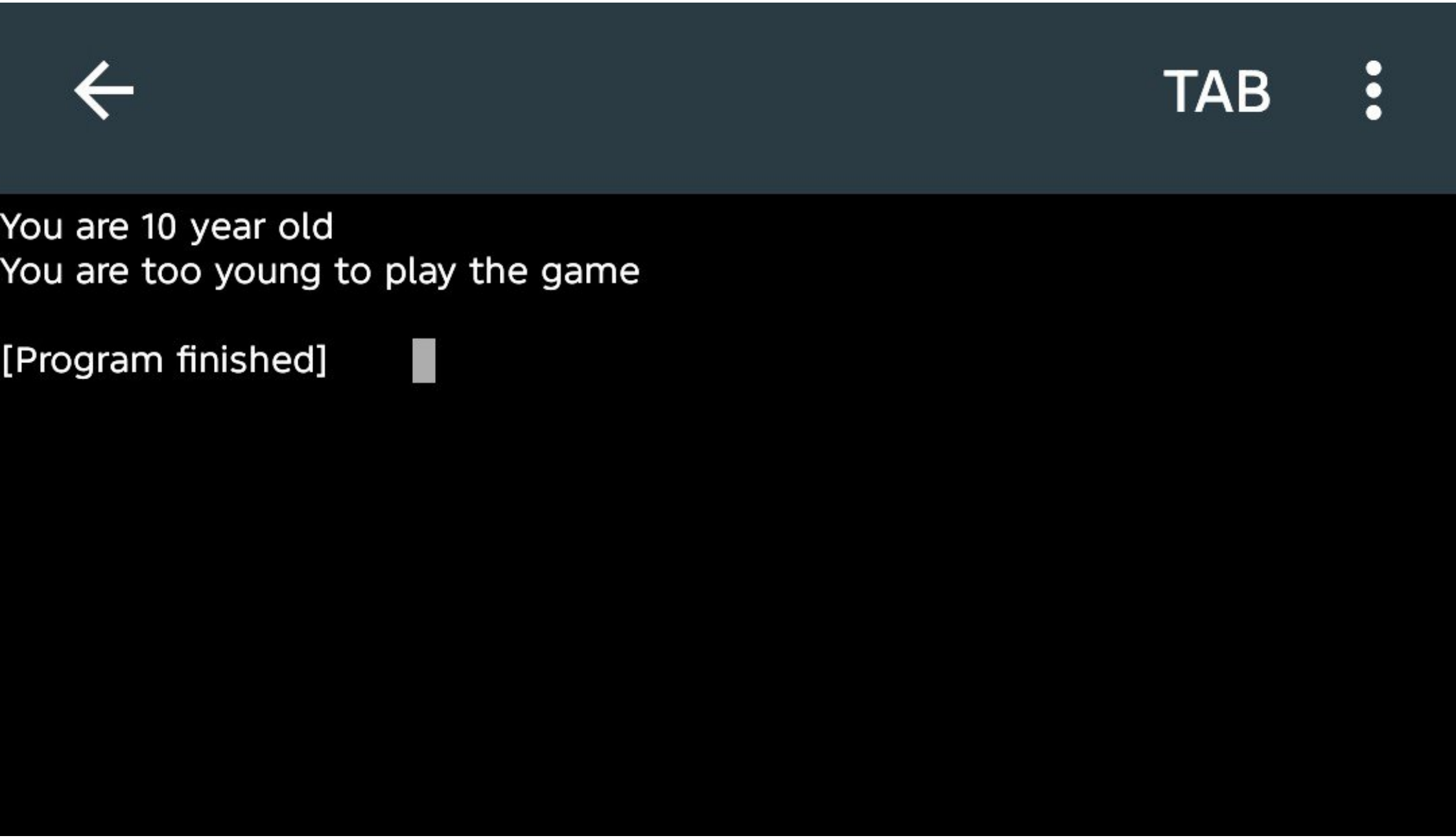
write a c++ program to print the following lines:

you are 10 year old
you are too young to play the game

```
#include <iostream>
using namespace std;
int main()
{
  cout << "you are 10 year old"<<endl;
  cout <<"you are too young to play the game"<<endl;
  System ("Pause");
  return 0;
}
```

▶ Output

you are 10 year old
you are too young to play the game


TAB
You are 10 year old
You are too young to play the game
[Program finished]

Day:

P-F- LAb

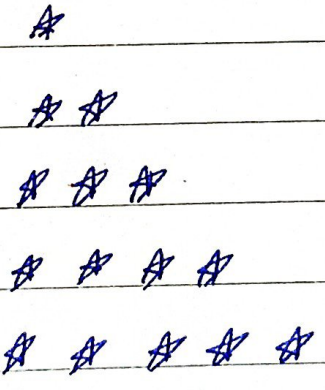
Date:

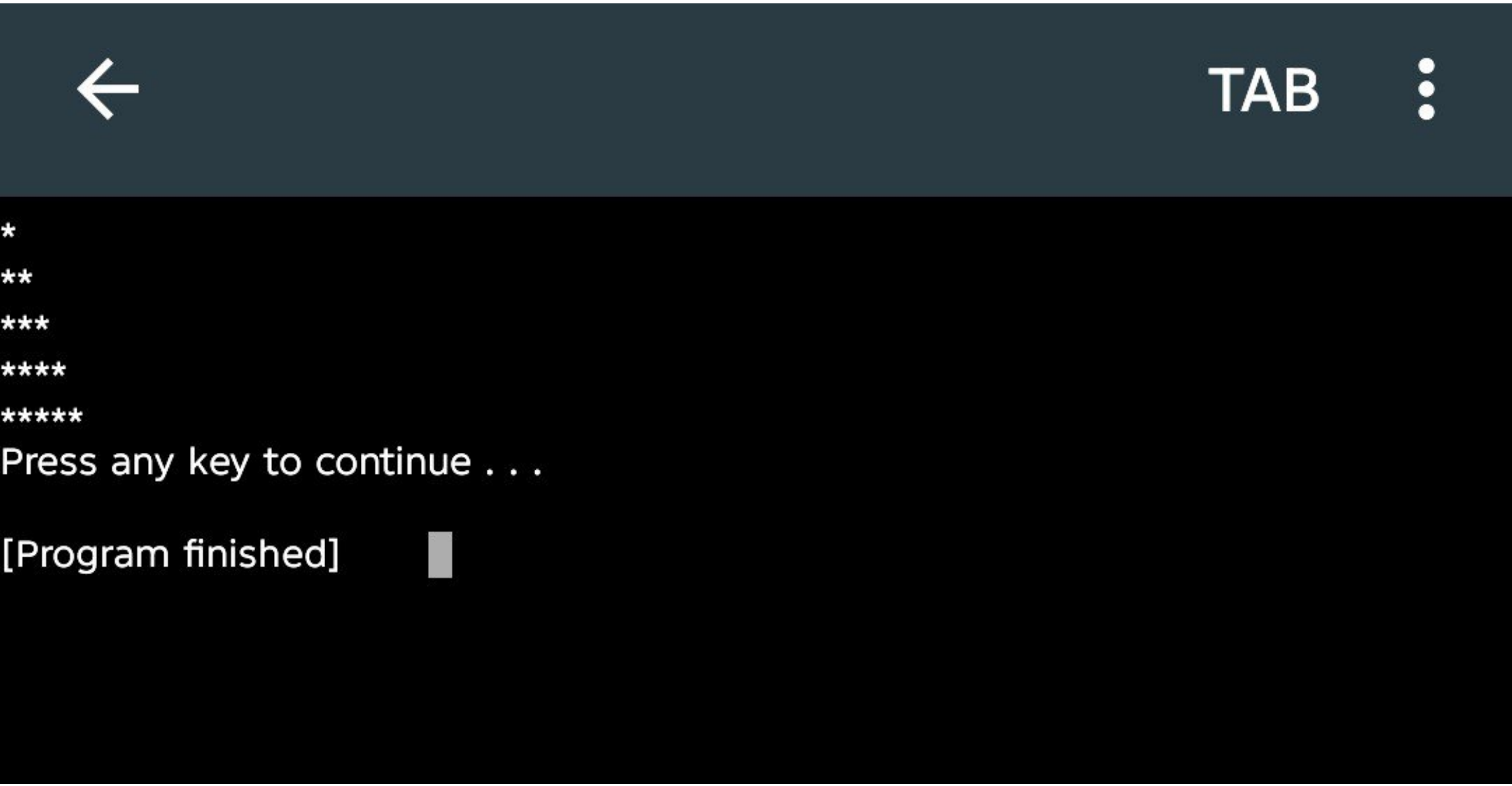
Task-3

[Star pattern]

```
# include <iostream>
using namespace std;
int main()
{
 cout << "*" << endl;
 cout << "**" << endl;
 cout << "***" << endl;
 cout << "****" << endl;
 cout << "*****" << endl;
 system ("pause");
 return 0;
 }
```

→ output




TAB
*
**
***
****
*****
Press any key to continue . . .
[Program finished]

Day: Date:

PF- LAb – 2

Task 1

Run these Programs & observe output

Program 1

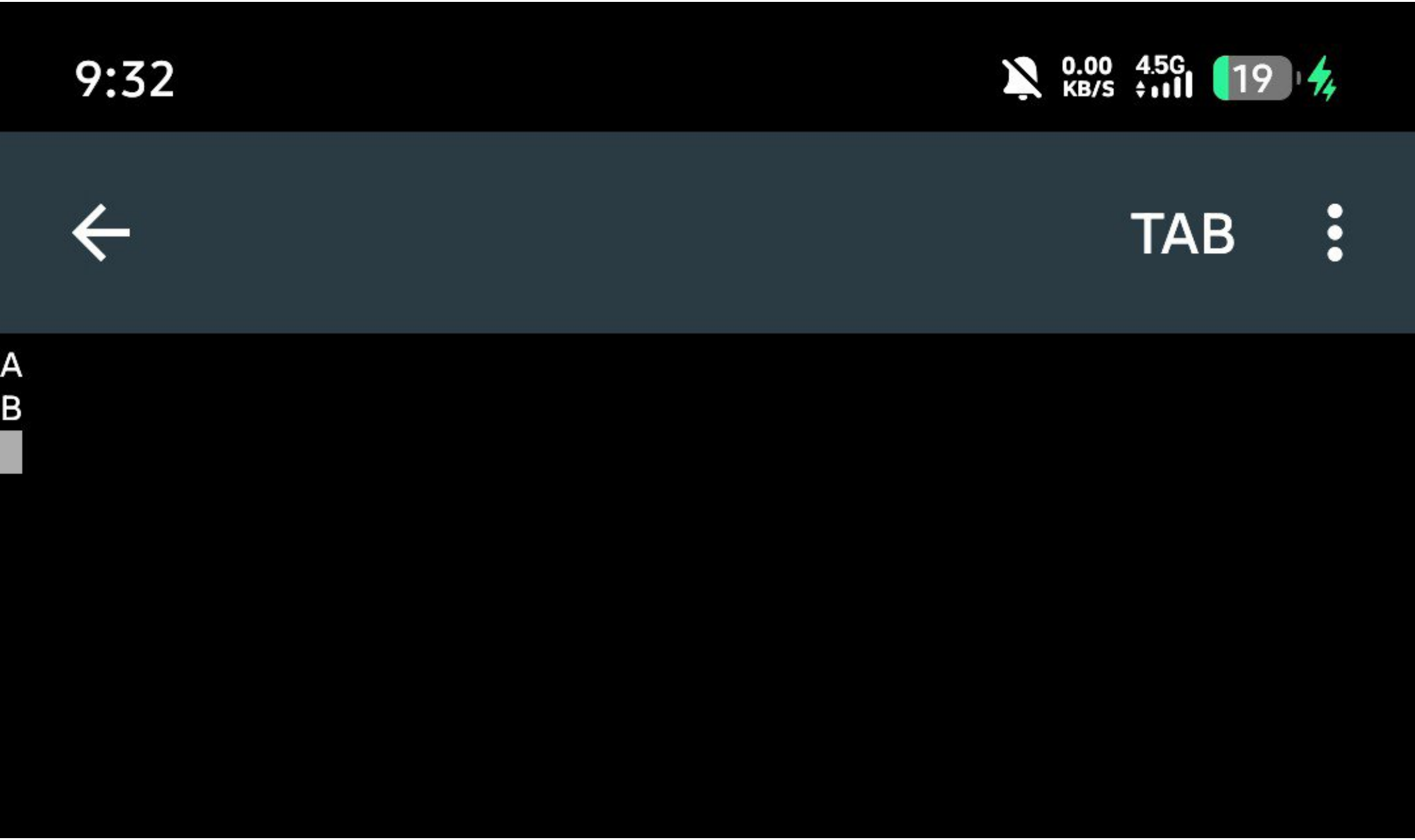
[input]

```
#include <iostream>
using namespace std;
int main()
{
    char Letter;
    Letter = 'A';
    cout << Letter << '\n';
    Letter = 'B';
    cout << Letter << '\n';
    cin >> Letter;
    return 0;
    }
```

[Output]

A

B


9:32
0.00
KB/S
4.5G
19
TAB
A
B

Day: _____ Date: _____

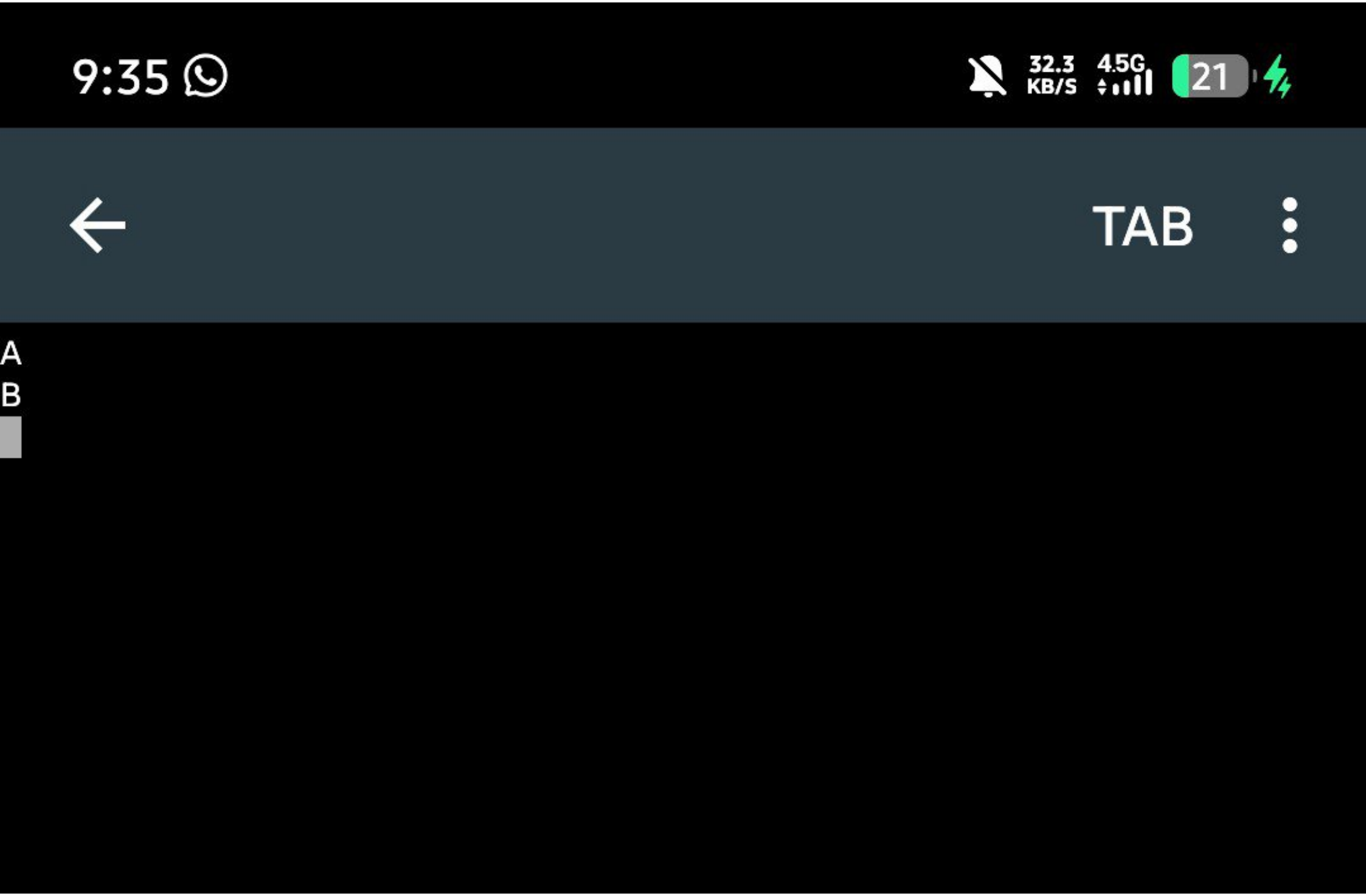
Program-2

[input]

```
#include <iostream>
using namespace std;
int main()
{
  char letter;
  letter = 65;
  cout << lecture << '\n';
  Letter = 66;
  cout << letter << '\n';
  cin >> letter;
  return 0;
  }
```

[output]

```
A
B
```


9:35
32.3 KB/S
4.5G
21
TAB
A
B

Day: _____ Date: _____

Program - 3

[input]

```
# include <iostream>
using namespace std;
int main ()
{
   int letter;
   letter = 'A';
   cout << letter << '\n';
   letter = 'B';
   cout << letter << '\n';
   cin >> letter;
   return 0;
   }
```

[output]

```
65
66
```


TAB
65
66

Day: _____ Date: _____

Task - 2

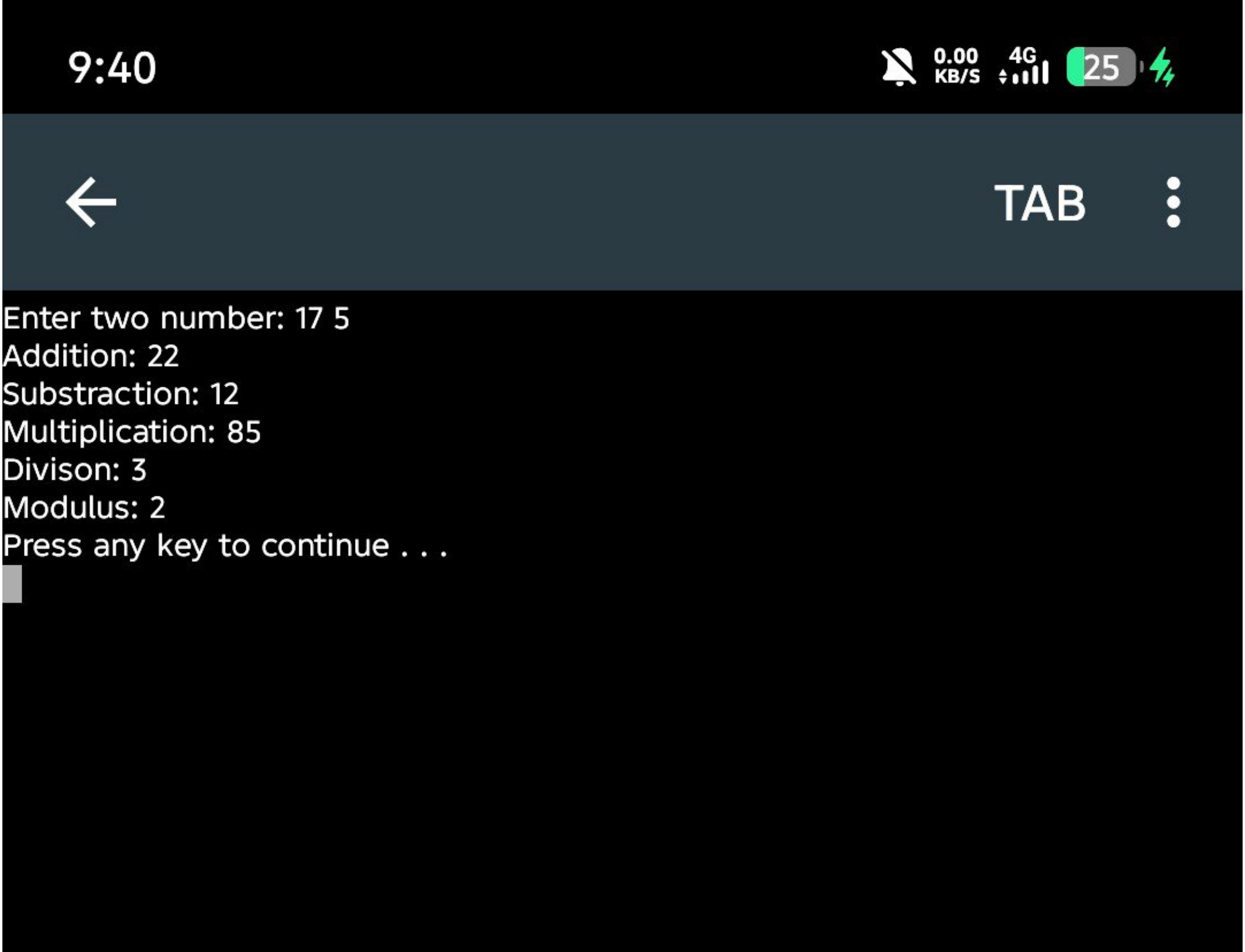
Take two integers as input Perform arithmatic operations on them

```
#include <iostream>
using namespace std;
int main()
{
int a, b;
cout << "Enter two number:";
cin >> a >> b;
cout << "Addition:" << a+b << endl;
cout << "Substraction:" << a-b << endl;
cout << "Multiplication:" << a*b << endl;
cout << "Divison:" << a/b << endl;
cout << "Modulus:" << a%b << endl;
System ("Pause");
return 0;
}
```

output (if a = 17, b = 5)

Enter two number 17 5

Addition: 22
Substraction: 12
Multiplication: 85
Divison: 3
Modules : 2


9:40
0.00
KB/S
4G
25
TAB
Enter two number: 17 5
Addition: 22
Substraction: 12
Multiplication: 85
Divison: 3
Modulus: 2
Press any key to continue . . .

Day: Date:

Task - 3

write output of following code

```
#include "Stdafx.h"
# include <iostream>
using namespace std;
int main ()
{
 int val1;
 int val2;
 int val3;
 cout << "Please enter your 3 numbers:";
 cin >> val1 >> val2 >> val3;
 cout <<val1<< "\n" << val2 << "\n" <<
        val3 << "\n";
 cout << val3 << "\n" << val2 << "\n" <<
        val1 << "\n";
 return 0;
}
```

output (if input is 1 2 3)

1
2
3
3
2
1


TAB
Please enter your 3 numbers: 1 2 3
1
2
3
3
2
1
[Program finished]

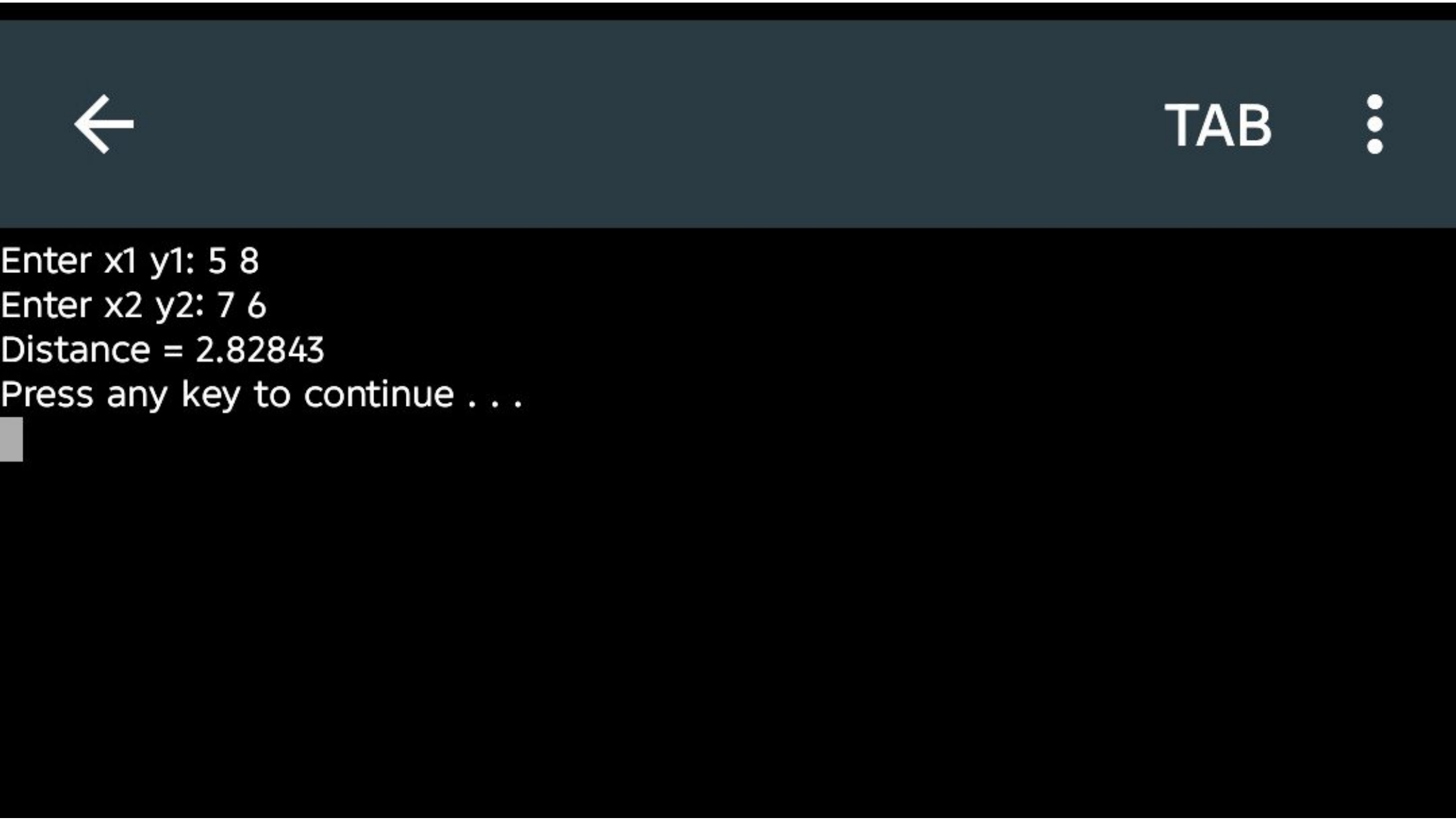
Day: Date:

Task - 4

Write c++ Program to calculate distance b/w two points. note x_1, y_1, x_2, y_2 are all double values

```
#include <iostream>
#include <cmath
using namespace std;
int main()
{
 double x1, y1, x2, y2, d;
 cout << "Enter x1 y1: ";
 cin >> x1 >> y1;
 cout << "Enter x2 y2: ";
 cin >> x2 >> y2;
 d = Sqrt (Pow(x2 - x1, 2) + Pow(y2 - y1, 2));
 cout << "Distance = " << d << endl;
 System ("Pause");
 return 0;
}
```

```
Enter x1 y1: 5 8
Enter x2 y2: 7 6
Distance: 2.82843
```


TAB
Enter x1 y1: 5 8
Enter x2 y2: 7 6
Distance = 2.82843
Press any key to continue . . .

Day: Date:

Task - 5

find circle area, rectangle area, Triangle area.

```
#include <iostream>
using namespace std;
int main()
{
  const double PI = 3.14159;
  double r, length, width, base, height;
  cout << "Enter radius :";
  cin >> r;
  cout << "Enter length and width :";
  cin >> length >> width;
  cout << "Enter base and height:";
  cin >> base >> height;
  cout << "circle area =" << PI*r*r << endl;
  cout << "Rectangle area =" << lengthxwidth
                             << endL;
  cout << "Triangle area =" << basexheight x0.5
                            << endL;
  System ("Pause");
  return 0;
}
```

circle (R)
∵ a = l×w
∵ Rect(i)
a = PI×r²
∵ Tri
are = b×h×5

```
Enter radius : 5 6
Enter Length and width : 7 8
```

Day: _____ Date: _____

Enter base and height 8 9

circle area = 78.5397

Rectangle area= 72

Triangle area= 32



TAB



```
Enter radius: 5 6
Enter length and width: 7 8
Enter base and height: 8 9
circle area = 78.5397
Rectangle area = 42
Triangle area = 32
Press any key to continue . . .
```

Task 6

Total Purchase. A customer in a store is purchasing five items. Price is given.

price of item 1 = $ 12.95
price of item 2 = $ 24.95
price of item 3 = $ 6.95
price of item 4 = $ 14.95
price of items = $ 3.95

write a program that holds the prices of the five items in five variables. Display each item price, the subtotal of the sale. the amount of sale, tax and total. Assume the sales tax is 6%.

Day: Date:

```
#include <iostream>
using namespace std;
int main()
{
    double p1 = 12.95, p2 = 27.95, p3 = 6.95, p4 = 17.95
           p5 = 3.95;
    double Subtotal = p1+p2+p3+p4+p5;
    double tax = Subtotal * 0.06;
    double total = Subtotal + tax;
    cout << "item 1: $" << p1 << endl;
    cout << "item 2: $" << p2 << endl;
    cout << "item 3: $" << p3 << endl;
    cout << "item 4: $" << p4 << endl;
    cout << "item 5: $" << p5 << endl;
    cout << "Subtotal: $" << Subtotal << endl;
    cout << "Sales Tax (6%): $" << tax << endl;
    cout << "Total: $" << total << endl;
    System ("Pause");
    return 0;
}
```

```
item 1: $12.95
item 2: $ 27.95
item 3: $ 6.95
item 4: $ 17.95
item 5: $  3.95
Subtotal: $ 63.75
```

```
Sales Tax(6%): 3.82$
Total: $ 67.57$
```



TAB



```
Price of item 1 = $ 12.95
Price of item 2 = $ 27.95
Price of item 3 = $ 6.95
Price of item 4 = $ 14.95
Price of item 5 = $ 3.95
Subtotal = $ 66.75
Sales Tax (6%) = $ 4.005
Total = $ 70.755
Press any key to continue . . .
```

Day: Date:

Bonus task

Write a program that inputs a five-digit integer, seperate the integer into individual digit and prints the digits space from each other by three space each.

```
#include <iostream>
using namespace std;
int main ()
{
  int n;
  cout << "Enter a 5-digit number:";
  cin >> n;
  cout <<n/10000 << "  "
      << (n/1000) %10<< "  "
      << (n/100) %10 << "  "
      << (n/10) %10<< "  "
      << n% << endl;
      System ("Pause");
      return 0;
    }
```

Enter a 5-digit number 24145

2 4 1 4 5



TAB



```
Enter a 5-digit number: 24145
2 4 1 4 5
sh: Pause: not found

[Program finished]
```

Day: Date:

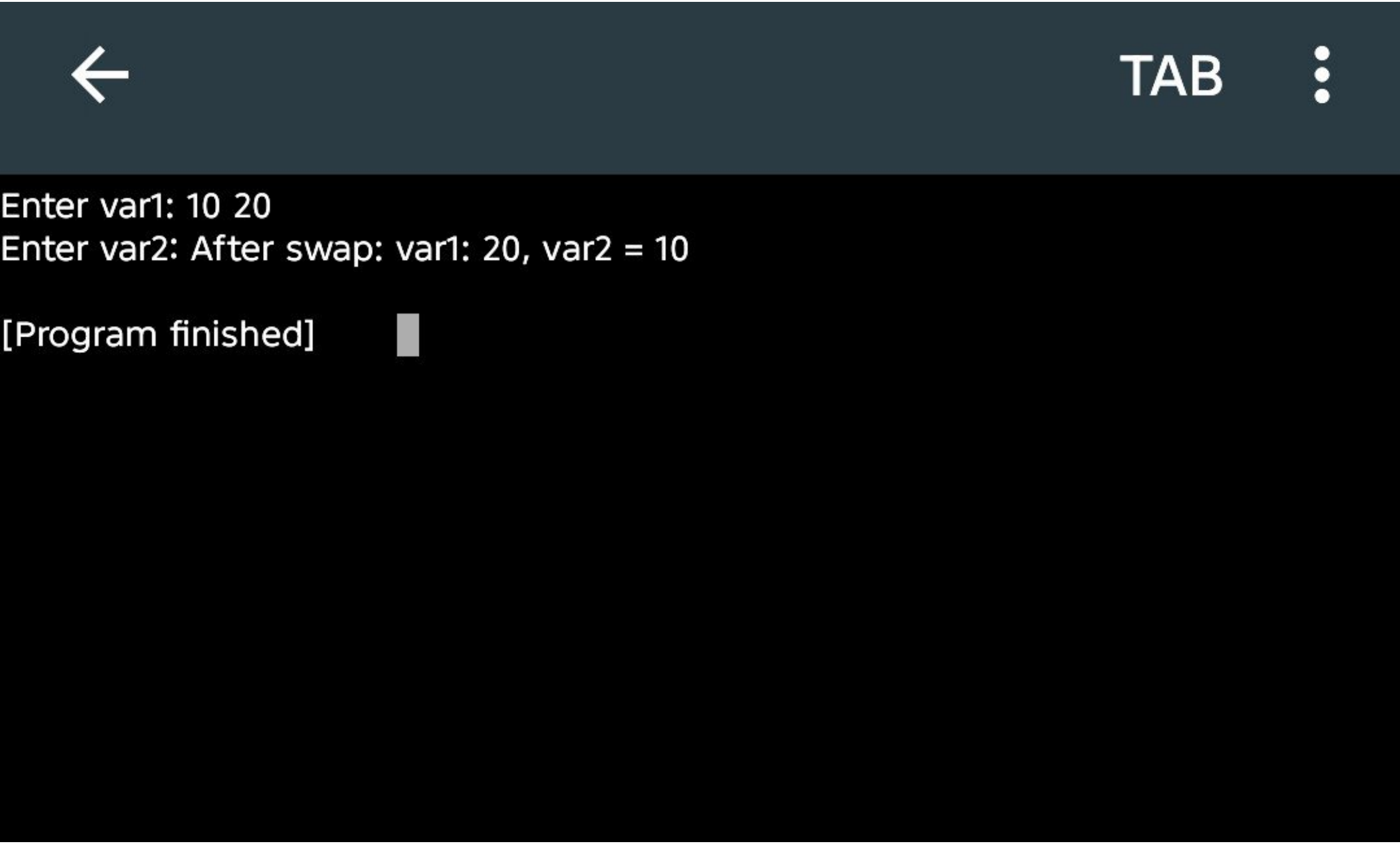
Home task 1

Swap

```
# include <iostream>
using namespace std;
int main ()
{
  int var1, var2, temp;
  cout << "Enter var1:";
  cin >> var1;
  cout << "Enter var2:";
  cin >> var2;
  temp = var1;
  var1 = var2;
  var2 =  temp;
  cout << "After Swap: var1: "<<var1<<";
          var2 = "<< var2<< endl;
  System ("Pause");
  return 0;
```

output

```
Enter var1:10
Enter var2:20
After Swap: var1=20, var2 : 10
```


TAB
Enter var1: 10 20
Enter var2: After swap: var1: 20, var2 = 10
[Program finished]

Day:

Date:

Home task (2)

Swap two variable

```
#include <iostream>
using namespace std;
int main()
{
  char letter, upper;
  cout << "Enter an alphabet in Small cap
                    : (a-z): ";
  cin >> letter;
  upper = letter - 32;
  cout << " upper case = " << upper << endl;
  System ("Pause");
  return 0;
}
```

```
Enter an alphabet in Small caps(a-z):
a b c d
upper case: A
```



TAB



```
Enter an alphabet in small case (a-z): z
upper case = Z

[Program finished]
```